

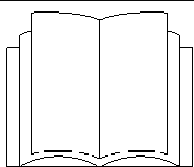


Operator's manual for Attachment



Log splitter

Product number:	300-700-series loaders:	A2951 A21278
	200-series loaders:	A21117



Please read this manual carefully before using the equipment, and follow all instructions.

Keep this manual for later reference.

Manufacturer:

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This manual uses the following warning symbols to indicate factors that must be taken into account to reduce the risk of personal injury or property damage:

	DANGER:
	This safety symbol refers to important safety information in this manual. It warns of an immediate hazard that could cause serious personal injury or property damage. Read the warning text accompanying the symbol carefully and ensure that other operators are also familiar with the warnings, since personal safety is at stake.



Note:

This notice symbol indicates information about the correct operation and maintenance of the equipment.

Failure to observe the instructions accompanying the symbol can lead to equipment breakdown or other property damage.

In addition to the safety instructions included in this manual, you must observe all occupational safety regulations, local laws and other regulations concerning the use of the equipment. Due to continuous product development, some of the details in this manual may differ from your equipment. We reserve the right to change the contents of the manual without notification. This manual is a translation of the original Finnish version.

1. Foreword

Avant Tecno Oy would like to thank you for your purchase of this AVANT attachment. It has been designed and manufactured on the basis of years of product development and experience. By familiarising yourself with this manual and following the instructions, you ascertain your safety and ensure the reliable operation and long service life of the equipment. Read the instructions carefully before starting to use the equipment or performing maintenance.

With these instructions, a person with basic knowledge and experience about the loader can use the attachment safely. The manual includes important instructions for experienced AVANT operators as well. Ensure that all persons using the loader have received proper guidance and familiarised themselves with the manual of the loader and attachment and all safety instructions before using the equipment. Keep this manual at hand throughout the service life of the equipment. If the manual is lost or damaged, you can request a new one from the retailer or manufacturer. If you sell or transfer the equipment, be sure to hand over this manual to the new owner.

2. Designed purpose of use

The AVANT log splitters are attachments designed and manufactured to be suitable for use with AVANT multi purpose loaders. They are intended to be used by one person at a time for splitting firewood whose diameter is max 400 mm. The splitting is done by a hydraulic cylinder, which presses the wood against a splitting wedge. The height of the splitting wedge can be adjusted in a way that the wood will be split into halves instead of quarters. The splitter returns the cylinder automatically to the initial position after the log has been split. Both the starting and ending points of the cylinder are adjustable according to the length of the wood. During splitting operation the area near the log, the splitting wedge and the cylinder is protected by a protective grille.

The attachments have been designed to require as little maintenance as possible. Regular maintenance tasks can be performed by the operator. Demanding repair and maintenance operations, such as hydraulic repairs, are to be left for professional maintenance. Familiarise yourself with the manual's instructions regarding service and maintenance. Please contact your AVANT retailer if you have additional questions about the operation or maintenance of the equipment or if you require spare parts or maintenance services.

3. Safety instructions for the log splitter

Please bear in mind that safety is the result of several factors. The loader-attachment combination is highly powerful, and improper or careless use or maintenance may cause serious personal injury or property damage. Due to this, you should carefully familiarise yourself with the manuals of both the loader and the attachment before starting operation. Do not use the attachment if you have not completely familiarised yourself with its operation and the related hazards. The purpose of these instructions is to help you to:

- use the equipment in a safe and efficient manner
 - observe and prevent any hazardous situations
 - keep the equipment intact and ensure a long service life
-
- When mounting the log splitter to the loader, ensure that the **locking pins** of the loader's quick attach plate are **in the lower position** and that they have **locked** the attachment **to the loader**.
 - The log splitter is to be used by one operator at a time. The operation of the log splitter creates a crushing hazard; do not let others near the machine when it is in use. Keep a safety distance of minimum of 2 meters.
 - When using the log splitter, keep it securely against the ground and attached to the loader. Read the instructions about the work site from section 6.
 - Always transport the log splitter as low and close to the machine as possible to keep the centre of gravity low.





- Pay attention to the contours of the terrain as well as other hazards, such as branches and trees reaching the driver's area. Pay attention to the surroundings and any other persons and machines moving in the vicinity.
- Shut down the loader engine before any maintenance or adjustments.
- Do not modify the structure of the attachment in a manner that would affect its safety.
- Do not use the log splitter to other than its intended use.
- Also read the safety instructions for the loader in the loader manual.



Familiarise yourself with the loader controls in a safe area. Pay particular attention to the safe stopping of the attachment and the machine.



Note that the loader boom moves down and may be tilted even if the engine has been stopped. Never go near the raised boom or the attachment.

Remember to wear proper protective clothing:



Never wear loose clothing, long scarves, jewellery or other items that might become entangled with the moving or rotating parts of the splitter or the wood being split. If needed, also wear a hat to prevent long hair from coming into contact with the splitter.



- The noise level in the driver's seat and the operating area may exceed 85 dB(A). Wear hearing protection while working with the loader.



- Wear safety boots while working with the log splitter.
- Wear protective gloves when handling logs or attaching the hydraulic hoses.

4. Technical specifications

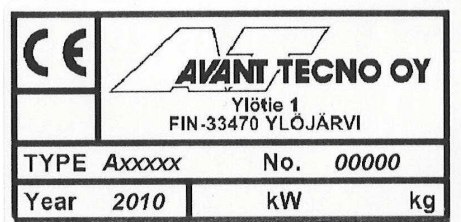
Table 1 – Log splitter technical data specifications

Product number	A21117	A2951	A21278
Log length, max:	600mm		1000 mm
Log diameter, max:	400 mm		
Weight:	150 kg		250 kg
Type of splitting wedge:	Splits to 2 or 4 sections		
Feeding speed:	Appr. 40 logs/min		
Maximum input of hydraulic energy:	60 l/min, 22,5 MPa (225 bar)		
Suitable AVANT loader:	200-sarja	300-700	300-700

Shown on the following page are the labels and markings, which must be visible on the equipment. Replace the warning labels, if they have become unclear or detached completely. New labels are available via your retailer or contact information provided on the cover.



The warning labels contain important safety information. Replace damaged or missing warning labels with new ones.



Identification plate of the attachment

- A46771 – Read instructions before use.
- A46772 – Do not go under a raised attachment, stay away from raised equipment.
- A46803 – Crushing hazard, stay clear of moving parts.
- A46749 – Shearing hazard, keep hands away from the splitting tools



A46771



A46772



A46803



A46749

5. Assembly of the log splitter

The log splitters are ready for use as delivered. Mount the attachment and its hydraulic hoses according to the instructions and read the following instructions for operating, adjustments and maintenance of the log splitter before starting to use the equipment.



Never move or raise an attachment that has not been locked. Keep the log splitter that has not been attached at even and firm ground.

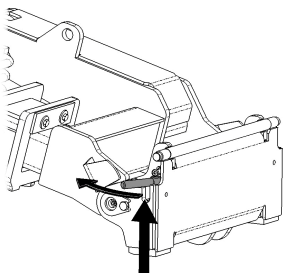
Mounting the attachment to the loader is quick and easy, but it must be done carefully. If the attachment is not locked to the loader, it may detach from the loader and cause hazardous situations. In the worst case, the attachment may slide onto the driver along a raised boom. For this reason, the attachment must under no circumstances be raised over one metre when it has not been locked. To prevent hazardous situations, always follow the attachment mounting instructions provided in the following pages. Also remember the safety instructions in Section 3.

5.1. Mounting the log splitter to the loader

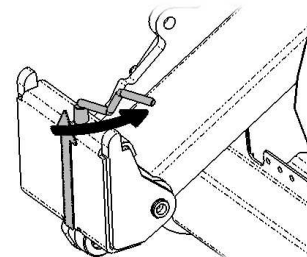
If your machine is equipped with the hydraulic locking and/or serial quick attach system, refer to the relevant manuals for additional mounting instructions.

The attachment is mounted to the AVANT loader boom quick attach plate and the counterpart on the attachment. The locking is semi-automatic, but because the attachment plate of the log splitter is at relatively high level, the locking pins must be locked manually. The attachment is mounted to the loader as follows:

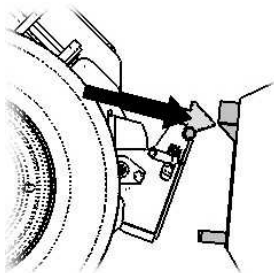
Step 1:



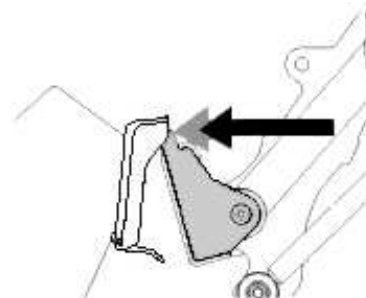
- Lift the quick attach plate locking pin / locking pins up and turn them backwards into the slot so that they are locked in the upper position.
- Ensure that the hydraulic hoses are not in the way during installation.



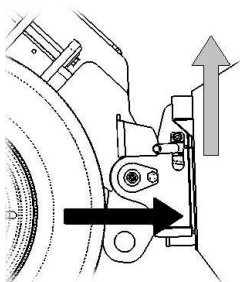
Step 2:



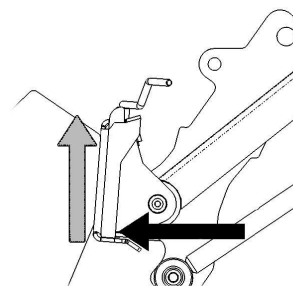
- Turn the quick attach plate hydraulically to an obliquely forward position.
- Drive the loader onto the attachment. If your loader is equipped with a telescopic boom, you can utilise this.



Step 3:



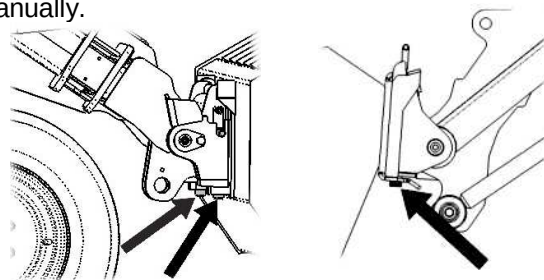
- Lift the boom slightly – pull the boom control lever backward to raise the attachment off the ground.
- Turn the boom control lever left to turn the bottom section of the quick attach plate onto the attachment.
- Lock the locking pins manually as you connect the hydraulic fittings.



Do not use the automatic return of the locking pins with the log splitter. Because the attachment plate of the log splitter is at relatively high level, tilting of an unlocked splitter increases the risk of toppling over. We recommend that you lower the log splitter onto the ground and secure the locking manually.



Before moving or raising the attachment, ensure that the locking pins are in the lower position and come through the fasteners on both sides.

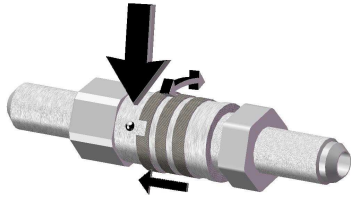


5.2. Connecting and disconnecting the hydraulic fittings

The log splitter has two hydraulic fittings, which must be connected to the corresponding fittings on the loader. When connecting or disconnecting fittings, shut down the loader engine and move the auxiliary hydraulics lever of the loader to its extreme positions a few times. This way, there will be no residual pressure in the system and the fittings are easy to connect.

To connect and disconnect the fittings, move the collar at the end of the female fitting. For correct operation of the splitter, connect the fittings with the coloured caps first, and the remaining fittings will then be fitted correctly.

In a 200-series equipment, the hose can be connected by pushing the fitting in the counterpart. When connecting and disconnecting the 200-series hydraulics quick couplings, you should pay attention to their locking. The locking prevents the female and male fittings from being disconnected from each other by accident when the attachment moves. The fitting collar may also turn by accident into the locking position, making it impossible to connect or disconnect the fittings.



In the 200-series equipment, there is a small groove in the collar of all female fittings and a small ball below the collar. When the fittings need to be disconnected, the above-mentioned groove and ball need to be aligned. Then, the grooved collar of the female fitting is pulled downwards to disconnect the fittings from each other.



When taking the attachment off from the loader, always disconnect the hydraulic fittings before detaching the quick attach plate, to prevent hose damage and any oil spills. Reinstall the protective caps on the fittings to prevent impurities from entering the hydraulic system.



Never connect or disconnect quick couplings or other hydraulic components while the system is under pressure. High-pressure fluid injection may cause serious injuries.

Keep the fittings as clean as possible; use the protective caps for quick couplings on the attachment and loader. Dirt, ice, etc. may make using the fittings significantly more difficult. Note that the protective caps on the loader and the attachment can be fastened to each other during operation to reduce the accumulation of dirt. Never leave the hoses hanging on the ground; place them on top of the equipment.

6. Operation

Once the log splitter has been mounted according to the instructions and the hydraulic hoses are connected, it is ready for work. Read the following instructions about work site, adjustments and use of the log splitter.

6.1. Setting up the log splitter

- Transport the log splitter carefully and note the effect of weight on the front to the stability of the loader. The splitter extension plate must be removed or folded during transportation.
- Choose a level working site, which can carry the load of the equipment and the logs. Clear the area from any obstacles that could cause a tripping hazard. The working area must not be slippery.
- Operate the log splitter in daylight or good artificial light and only in well-ventilated areas.
- Lower the splitter on the ground and apply the parking brake of the loader.



Lower the log splitter firmly on the ground for use. Never go under a raised attachment or the loader boom. Note that the loader boom may be lowered or tilted even if the engine has been shut down.



Never perform any adjustments, maintenance or cleaning of the equipment if the splitter can start by accident. Shut down the loader engine first.



Never leave the equipment unattended while the engine is running.



Shut down the loader engine and release the residual pressure to prevent the splitting cylinder from moving during adjustments. If there are other persons at the work area also remove the keys from the ignition or disconnect a hydraulic fitting.

6.1.1. Adjusting the stroke length

When the logs are shorter than the cylinder stroke, the stroke can be adjusted so that unnecessary travel of the cylinder is minimized, thus speeding up the work. Both the starting and the stopping position of the carriage can be adjusted.

The stroke length is adjusted by moving the adjustment screws on the push rod. By moving the right stopper to the left the cylinder will not return to its original starting point. By moving the left stopper to the right, the cylinder will start its return earlier.



Stopper screws for adjusting the stroke length



The stroke must be adjusted so that the hydraulic cylinder (carriage) stops at least 10 cm before the end of stroke. In most cases this will be enough for splitting the wood.

6.1.2. Adjusting the wedge height

The height of the splitting wedge can be adjusted to match the log diameter. It can also be set to split the log to two pieces. The wedge height is adjusted as follows:

1. If the log splitter is mounted on the loader, shut down the loader and release the residual pressure to make sure that the cylinder will not move unexpectedly during the adjustment.
2. Lift the protective grille up.
3. Remove the cotter pin from the pin shown in adjacent figure and support the wedge by hand. Pull the pin out.
4. When the pin has been removed, the wedge can be lifted or lowered as needed. Place the pin back and secure it with the cotter pin. If you want to split the wood to only two pieces, lower the wedge to its lowest position.



6.2. Use

Turn the auxiliary hydraulics control lever on the loader to its locked position. The control device is a lever that locks to its front or side position or a pedal that locks to its lower position.

Lower the attachment firmly on the ground, apply the parking brake of the loader and set moderate engine revolutions depending on the working pace. Depending on the loader model, two hydraulic pumps may be used to reduce engine noise. Note that unnecessarily high engine rotating speed will not improve the splitting force.

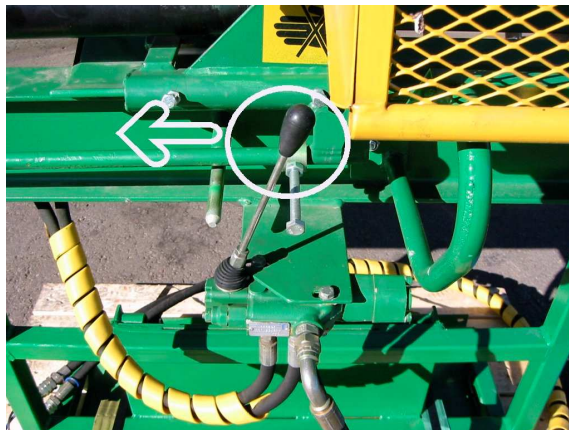
- Check that the log splitter is intact and that all obstacles have been removed from the working area before starting the work.
- Check the correct operation of the controls by completing a few empty cycles with the splitter. The splitter should perform one splitting cycle only if the protective grille is closed, and after the cycle the cylinder should automatically return to its initial position and stop.
- Operate the attachment only from the operator's area next to the controls of the splitter.
- Ensure that the loader and the splitter are used in a safe and intended manner. Smoking near the operating area is prohibited. Do not allow children to operate the equipment.
- Check regularly to ensure there are no hydraulic leaks. Refer to Section 7 for maintenance instructions.



Never reach to the area protected by the grille while the loader engine is running. Before using the log splitter check that the grille is in place and that it prevents the splitting operation when it is opened. Do not operate the splitter without the guard in place.

Load a log against the splitting carriage, so that the squarer end of the log is placed towards the plate. During the splitting the protective grille must remain closed. Never attempt to split more than one log at a time unless the carriage has been fully extended and a second log is needed to complete the separation of the first log.

The splitting cylinder will start its splitting cycle when the control lever (shown below) is turned to the left. The cylinder will complete one full splitting cycle, as limited by the stoppers shown in section 6.1.1 and the cylinder will stop to its initial position. **The cylinder can be stopped by turning the control lever to its middle position.** The cylinder can be returned to its initial position manually by turning the lever to the right.



The control lever of the log splitter



In case the log gets stuck or other disruptions in use, first turn the control lever to its middle position and shut down the loader engine before other procedures. Solve the disruption after the engine has been shut down.



Always keep your fingers away from any cracks that open in the log during splitting. They can quickly close and pinch your fingers.



The log splitter is intended to be used by one operator only.

7. Maintenance and servicing

The log splitter does not require scheduled maintenance. Keep the equipment clean and monitor the condition of the hydraulic components.

You may use a pressure washer and a mild detergent to wash the equipment. Avoid spraying directly at the hydraulic parts at high pressure. In addition, mind the labels on the attachment. Do not use solvents when washing.

Clean the attachment before long storage periods, touch up the paint where needed and grease the lubrication points and the chain drive to prevent rust damage. The cylinder should be completely retracted during storage. Apply a small amount of oil to the visible part of the cylinder shaft.

Check the condition of the hydraulic hoses and components when the engine has been turned off and the pressure has been relieved. Repair all leaks immediately after detecting them; a small leak can quickly grow into a big one. Do not use the equipment, if you have discovered a leak in the hydraulic system. Leaking hydraulic fluid may penetrate skin and cause serious injuries. The fluid is also harmful to the environment.

Check hoses visually for cracks or abrasions. Monitor the wearing of the hoses and stop the use, if the surface layer of the hose has worn off. If there are signs of leaking, to check the component, hold up a piece of cardboard in the area where a leak is suspected. Finding any of these faults means that the hydraulic hose or component must be replaced. Spare parts are available from your nearest AVANT retailer or service point.



Never tighten a hydraulic fitting under pressure, since the fitting may break and the released oil may cause serious injuries. Do not use the equipment, if you have discovered a fault in the hydraulic system.



8. Warranty terms

Avant Tecno Oy grants a warranty of one year (12 months) from the date of purchase for the attachment it manufactures.

The warranty covers repair costs as follows:

- Work costs are covered, if the repair is not performed at the factory.
- The factory replaces any defective components or consumables.

The factory may reimburse the price of components purchased by the customer in special cases that have been agreed in advance.

The warranty does not cover:

- Normal maintenance work or parts and consumables required for it.
- Damages caused by unusual operating conditions or ways of use, negligence, overheating, structural changes made without the consent of Avant Tecno Oy, use of non-original parts or lack of maintenance.
- Consequences of a defect, such as interruption of work or other possible additional damages.
- Travel and/or freight costs caused by the repair.

EY-vaatimustenmukaisuusvakuutus
EG-försäkran om överensstämmelse
EC Declaration of Conformity
EG-Konformitätserklärung



Valmistaja / Tillverkare / Manufacturer / Hersteller:

AVANT TECNO OY

Osoite / Adress / Address / Adresse:

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33470 YLÖJÄRVI, FINLAND

Vakuutamme täten, että alla mainitut tuotteet täyttävät konedirektiivin turvallisuus- ja terveysvaatimukset (direktiivi 2006/42/EY muutoksineen). Seuraavia yhdenmukaistettuja standardeja on sovellettu:
 SFS-EN ISO 12100-1, SFS-EN ISO 12100-2, SFS-EN 982, SFS-EN 609-1

Vi försäkrar härmed att nedan beskrivna produkter överensstämmer med hälso- och säkerhetskrav i EG-maskindirektiv (EG-direktiv 2006/42/EG som ändrat). Följande harmoniserade standarder har tillämpats:
 SFS-EN ISO 12100-1, SFS-EN ISO 12100-2, SFS-EN 982, SFS-EN 609-1

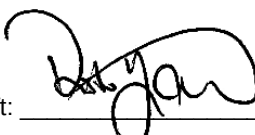
We hereby declare that the products listed below are in conformity with the provisions of the Machinery Directive (directive 2006/42/EC as amended). The following harmonized standards have been applied:
 SFS-EN ISO 12100-1, SFS-EN ISO 12100-2, SFS-EN 982, SFS-EN 609-1

Wir erklären hiermit, dass die nachstehend aufgeführte Maschine konform ist mit den Bestimmungen der EG-Maschinenrichtlinie (EG-Richtlinie 2006/42/EG mit Änderungen). Die folgende harmonisierte Normen wurden angewandt:
 SFS-EN ISO 12100-1, SFS-EN ISO 12100-2, SFS-EN 982, SFS-EN 609-1

Kuvaus: Avant-kuormainten kanssa käytettävä hydraulinen puunhalkaisija
 Beskrivning: Hydraulisk vedklyv; arbetsredskap för Avant lastare
 Description: Hydraulic log splitter; attachment for Avant loaders
 Beschreibung: Hydraulische Holzspalter; Anbaugerät für Avant Radlader

Mallit / Modeller / Models / Modellen:

Avant		200 series	300-700 series
Puunhalkaisija	60 cm	A21117	A2951
Vedklyv			
Log splitter	100 cm	-	A21278
Holzspalter			

Allekirjoitus / Namnteckning / Signature / Unterschrift:  10.6.2010 Ylöjärvi, Finland

Risto Käkelä,
 Toimitusjohtaja / Verkställande direktör /
 Managing Director / Geschäftsführer





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